MATHS MANTHAN
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 09** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM:** 30

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. If the arithmetic mean of $7, 8, x, 11, 14$ is x, then the value of x is: [1]

(A) 9 (B) 9.5 (C) 10 (D) 10.5

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 10

Justification:

$$\bar{x} = \frac{\sum x_i}{n} \implies x = \frac{7+8+x+11+14}{5}$$

$$5x = 40 + x \implies 4x = 40 \implies \boxed{x = 10}.$$

Q2. The difference between the upper limit of the median class and the lower limit of the modal class for the data is: [1]

Class	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25
Frequency	10	15	12	20	9

(A) 0 (B) 5 (C) 10 (D) 15

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 0

Justification: Total frequency $N = 10 + 15 + 12 + 20 + 9 = 66 \implies \frac{N}{2} = 33$.
Cumulative frequencies: $0 - 5 : 10$; $5 - 10 : 25$; $10 - 15 : 37$; $15 - 20 : 57$; $20 - 25 : 66$.
Median class is **10 – 15** $\implies$ Upper limit = **15**.
Maximum frequency is 20, which corresponds to modal class **15 – 20** $\implies$ Lower limit = **15**.

$$\text{Difference} = 15 - 15 = \mathbf{0}.$$

$$\boxed{\text{Difference} = \mathbf{0}}$$

Q3. If a variable takes values $0, 1, 2, \ldots, n$ with frequencies proportional to $1, 1, 1, \ldots, 1$, then its mean is: [1]

(A) $n/2$ (B) $(n+1)/2$ (C) $n(n+1)/2$ (D) n

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $n/2$

Justification: There are $n+1$ values: $0, 1, 2, \ldots, n$, each with equal frequency $f_i = 1$.

$$\text{Sum of values} = 0 + 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

$$\text{Total number of observations} = n + 1.$$

$$\text{Mean} = \frac{\frac{n(n+1)}{2}}{n+1} = \frac{\mathbf{n}}{\mathbf{2}}.$$

$$\boxed{\text{Mean} = \frac{\mathbf{n}}{\mathbf{2}}}$$

Q4. For a symmetrical frequency distribution, which of the following relations is correct? [1]

(A) Mean ¿ Median ¿ Mode (B) Mean ¡ Median ¡ Mode
(B) Mean = Median = Mode (D) Mode = $\frac{\text{Mean}+\text{Median}}{2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) Mean = Median = Mode

Justification: In a perfectly symmetrical frequency distribution (zero skewness), the mean, median, and mode coincide at the exact central axis: Mean = Median = Mode.

Q5. If the mode of a frequency distribution is 55 and the median is 50, then its mean is: [1]

(A) 47.5 (B) 48.0 (C) 52.5 (D) 45.0

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 47.5

Justification:

$$\text{Mode} = 3\text{Median} - 2\text{Mean} \implies 55 = 3(50) - 2\text{Mean}$$

$$55 = 150 - 2\text{Mean} \implies 2\text{Mean} = 150 - 55 = 95 \implies \boxed{\text{Mean} = \mathbf{47.5}}.$$

Q6. **Assertion (A):** The cumulative frequency of the last class interval in a 'less than' cumulative distribution is equal to the total frequency N.
Reason (R): Cumulative frequency of a class is the sum of frequencies of that class and all preceding classes. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: In a 'less than' distribution, frequencies accumulate from the lowest class to the highest class. Since the last class includes all observations, its cumulative frequency equals the total frequency N. Both statements are true and Reason (R) correctly explains Assertion (A).

SECTION B: Very Short Answer Type

MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. In a continuous distribution, $l = 40, h = 10, f_1 = 20, f_0 = 12, f_2 = 11$. Calculate the mode. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formula Declaration [1 Mark]

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

Step 2: Substitution & Calculation [1 Mark]

$$\text{Mode} = 40 + \left(\frac{20 - 12}{2(20) - 12 - 11}\right) \times 10 = 40 + \left(\frac{8}{40 - 23}\right) \times 10 = 40 + \frac{80}{17}$$

$$\text{Mode} = 40 + 4.706 \approx \mathbf{44.71} \quad \left(\text{or } 44\frac{12}{17}\right)$$

$$\boxed{\text{Mode} = \mathbf{44.71} \quad \left(\text{or } \mathbf{44\frac{12}{17}}\right)}$$

Q8. In a grouped distribution, $l = 25, N = 50, cf = 18, f = 14, h = 5$. Calculate the median. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formula Declaration [1 Mark]

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h$$

Step 2: Substitution & Calculation [1 Mark]

Here, $\frac{N}{2} = \frac{50}{2} = 25$.

$$\text{Median} = 25 + \left(\frac{25 - 18}{14}\right) \times 5 = 25 + \left(\frac{7}{14}\right) \times 5 = 25 + \frac{5}{2} = 25 + 2.5 = \mathbf{27.5}.$$

$$\boxed{\text{Median} = \mathbf{27.5}}$$

SECTION C: Short Answer Type

MATHS MANTHAN | $(2 \times 3 = 6$ Marks)

Q9. Find the median of the distribution of letters in surnames: [3]

Letters	1 – 4	4 – 7	7 – 10	10 – 13	13 – 16	16 – 19
Surnames	6	30	40	16	4	4

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Cumulative Frequency Table [1 Mark]

Number of Letters	Number of Surnames (f_i)	Cumulative Frequency (cf)
$1-4$	6	6
$4-7$	30	$6+30=36$
$7-10$	40	$36+40=76$
$10-13$	16	$76+16=92$
$13-16$	4	$92+4=96$
$16-19$	4	$96+4=100$
Total	$N=100$	

Step 2: Median Class & Parameters [1 Mark]

$N=100 \implies \dfrac{N}{2}=50.$

The cumulative frequency just greater than 50 is 76, corresponding to class **$7-10$**.

$\therefore$ Median Class $= 7-10$.

Lower limit $(l)=7$, $cf=36$, $f=40$, $h=3$.

Step 3: Median Calculation [1 Mark]

$$\text{Median} = l+\left(\frac{\frac{N}{2}-cf}{f}\right)\times h = 7+\left(\frac{50-36}{40}\right)\times 3 = 7+\frac{14\times 3}{40} = 7+\frac{42}{40} = 7+1.05 = \mathbf{8.05}$$

Median Number of Letters = **8.05**

Q10. The mean of the following distribution is 25. Find the value of p: [3]

Class	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$
Frequency	5	18	15	p	6

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Direct Method Calculation Table [1.5 Marks]

Class Interval	Frequency (f_i)	Class Mark (x_i)	f_ix_i
$0-10$	5	5	25
$10-20$	18	15	270
$20-30$	15	25	375
$30-40$	p	35	$35p$
$40-50$	6	45	270
Total	$\sum f_i = 44+p$		$\sum f_ix_i = 940+35p$

Step 2: Equation with Given Mean [1 Mark]

$$\bar{x}=\frac{\sum f_ix_i}{\sum f_i} \implies 25=\frac{940+35p}{44+p}$$

$$25(44+p)=940+35p \implies 1100+25p=940+35p$$

Step 3: Solving for p [0.5 Mark]

$$35p-25p=1100-940 \implies 10p=160 \implies p=16.$$

Value of p = **16**

SECTION D: Long Answer Type

MATHS MANTHAN | $(2\times 5=10$ Marks)

Q11. A life insurance agent found the following data for distribution of ages of 100 policy holders. Calculate the median age and the mean age: [5]

Age (in years)	Below 20	Below 25	Below 30	Below 35	Below 40
Number of policy holders	2	6	24	45	78

Age (in years)	Below 45	Below 50	Below 55	Below 60	–
Number of policy holders	89	92	98	100	–

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Frequency Distribution & Calculation Table [2 Marks]

Age Interval	**Frequency** (f_i)	**Cum. Freq.** (cf)	**Class Mark** (x_i)	$f_i x_i$
$15-20$	2	2	17.5	35.0
$20-25$	4	6	22.5	90.0
$25-30$	18	24	27.5	495.0
$30-35$	21	45	32.5	682.5
$35-40$	33	78	37.5	1237.5
$40-45$	11	89	42.5	467.5
$45-50$	3	92	47.5	142.5
$50-55$	6	98	52.5	315.0
$55-60$	2	100	57.5	115.0
Total	$\sum f_i = 100$			$\sum f_i x_i = 3580.0$

Step 2: Median Age Calculation [1.5 Marks]

$N = 100 \implies \dfrac{N}{2} = 50$. The median class is **35 – 40** ($l = 35, cf = 45, f = 33, h = 5$).

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h = 35 + \left(\frac{50-45}{33}\right) \times 5 = 35 + \frac{25}{33} \approx \mathbf{35.76} \text{ years} \quad \left(\text{or } 35\frac{25}{33} \text{ years}\right).$$

Step 3: Mean Age Calculation [1.5 Marks]

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{3580}{100} = \mathbf{35.8} \text{ years.}$$

Median Age = **35.76** years, Mean Age = **35.8** years

Q12. If the mean of the following frequency distribution is 145 and the total frequency is 100, find the missing frequencies f_1 and f_2: [5]

Class	$100-120$	$120-140$	$140-160$	$160-180$	$180-200$
Frequency	10	f_1	30	f_2	15

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Total Frequency Equation [1.5 Marks]

$$\sum f_i = 10 + f_1 + 30 + f_2 + 15 = 55 + f_1 + f_2 = 100$$

$$f_1 + f_2 = 100 - 55 \implies f_1 + f_2 = 45 \quad \text{— (i)}$$

Step 2: Direct Method Calculation Table [1.5 Marks]

Class	Frequency (f_i)	Class Mark (x_i)	$f_i x_i$
$100 - 120$	10	110	1100
$120 - 140$	f_1	130	$130f_1$
$140 - 160$	30	150	4500
$160 - 180$	f_2	170	$170f_2$
$180 - 200$	15	190	2850
Total	$\sum f_i = 100$		$\sum f_i x_i = 8450 + 130f_1 + 170f_2$

Step 3: Setting Up Mean Equation [1 Mark]

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \implies 145 = \frac{8450 + 130f_1 + 170f_2}{100}$$

$$14500 = 8450 + 130f_1 + 170f_2 \implies 130f_1 + 170f_2 = 6050$$

Dividing by 10:

$$13f_1 + 17f_2 = 605 \quad \text{— (ii)}$$

Step 4: Solving Simultaneous Equations [1 Mark]

From (i), $f_1 = 45 - f_2$. Substituting in (ii):

$$13(45 - f_2) + 17f_2 = 605 \implies 585 + 4f_2 = 605 \implies 4f_2 = 20 \implies f_2 = \mathbf{5}.$$

From (i): $f_1 = 45 - 5 = \mathbf{40}$.

$$\boxed{f_1 = \mathbf{40}, \quad f_2 = \mathbf{5}}$$

SECTION E: Case Study Based Question

MATHS MANTHAN | $(1 \times 4 = 4$ Marks$)$

Q13. CASE STUDY: NATIONAL EXPRESS HIGHWAY SMART FASTAG TOLL CLEARANCE EFFICIENCY [4 Marks]

The National Highways Authority analyzed vehicle transit processing delays (in minutes) at an interstate electronic toll plaza during holiday weekend congestion.

Transit Delay (minutes)	$0 - 5$	$5 - 10$	$10 - 15$	$15 - 20$	$20 - 25$
Vehicles Logged	15	35	50	30	20

(i) Calculate the modal delay time experienced by vehicles at the toll plaza. [2]

(ii) What is the total number of vehicles surveyed in this clearance audit? [1]

(iii) Find the median class of the delay distribution. [1]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Subpart (i): Modal Delay Time Calculation [2 Marks]

The highest frequency of vehicles is 50, corresponding to the interval **10 – 15**.
$\therefore$ Modal Class = 10 – 15 minutes.
$l = 10,\ h = 5,\ f_1 = 50,\ f_0 = 35,\ f_2 = 30.$

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \text{[1 Mark]}$$

$$\text{Mode} = 10 + \left(\frac{50 - 35}{2(50) - 35 - 30}\right) \times 5 = 10 + \left(\frac{15}{100 - 65}\right) \times 5 = 10 + \frac{75}{35} = 10 + \frac{15}{7}$$

$$\text{Mode} = 10 + 2.143 \approx \mathbf{12.14} \text{ minutes} \quad \left(\text{or } 12\frac{1}{7} \text{ min}\right) \text{[1 Mark]}$$

$$\boxed{\text{Modal Delay Time} = \mathbf{12.14} \text{ minutes} \quad \left(\text{or } \mathbf{12\frac{1}{7}} \text{ min}\right)}$$

Subpart (ii): Total Vehicles Surveyed [1 Mark]

$$N = 15 + 35 + 50 + 30 + 20 = \mathbf{150} \text{ vehicles.}$$

$$\boxed{\text{Total Vehicles Surveyed} = \mathbf{150}}$$

Subpart (iii): Median Class of the Distribution [1 Mark]

$N = 150 \implies \dfrac{N}{2} = 75.$
Cumulative frequencies: $0 - 5 : 15;\quad 5 - 10 : 50;\quad 10 - 15 : 100;\quad 15 - 20 : 130;\quad 20 - 25 : 150.$
The cumulative frequency just greater than 75 is 100, corresponding to class **10 – 15**.

$$\boxed{\text{Median Class} = \mathbf{10 - 15} \text{ minutes}}$$

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